uc3m | Universidad Carlos III de Madrid

Master Degree in Machine Learning for Health
2025-2026

Master Thesis

"Deep Learning for Photomultiplier Crystal Detection Signals"

Miguel Escudero Rodríguez

David Izquierdo-García

Leganés, September 2026

Deep Learning for Photomultiplier Crystal Detection Signals

Miguel Escudero Rodríguez
Master's Degree in Machine Learning for Health
Universidad Carlos III de Madrid, Leganés, Spain
Email: 100569281@alumnos.uc3m.es

***Abstract*—TODO: 150–250 palabras. Problema, metodo propuesto, datos empleados y resultados cuantitativos principales. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.**

Curabitur pretium tincidunt lacus. Nulla gravida orci a odio, sed fermentum lorem interdum. Phasellus convallis, sapien nec facilisis tincidunt, justo lectus malesuada erat, vitae consectetur neque arcu vel lorem. Integer consequat, augue a interdum consequat, erat sapien volutpat justo, at tincidunt libero mauris nec nisl. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Donec vitae augue sed justo interdum tincidunt.

Mauris euismod, lectus vitae tincidunt posuere, justo sem malesuada neque, eget feugiat sapien nisl a erat. Aenean interdum, purus vel malesuada consequat, magna sapien tincidunt augue, non consequat libero neque at justo. Suspendisse potenti. Proin vitae lorem at ipsum interdum tincidunt. Aliquam erat volutpat. Nam sed augue vel risus commodo interdum, sed tincidunt justo consequat. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Integer vel lorem nec sapien facilisis aliquam. Donec vulputate, magna at consequat suscipit, justo neque malesuada nisl, vitae tincidunt erat turpis vel ipsum. Vivamus feugiat lorem sed arcu malesuada, quis consequat purus tincidunt. Sed euismod, neque vitae interdum consequat, justo sapien fermentum mauris, vel tincidunt nulla nisl non erat.

***Index Terms*—Positron emission tomography, silicon photomultiplier, missing data imputation, deep learning, graph neural networks,silicon photomultiplier failure.**

I. Introduction

Positron Emission Tomography (PET) is a non-invasive molecular imaging modality and one of the central techniques of nuclear medicine, providing an *in vivo* assessment of the metabolic and biochemical state of a patient [1]. Unlike purely anatomical modalities, PET relies on radiotracers: biologically active molecules chemically labelled with positron-emitting radionuclides [2]. Once administered, the tracer distributes according to the physiological pathway targeted by its carrier molecule, so that the measured signal reflects function rather than structure. When an emitted positron annihilates with an electron in the surrounding tissue, two photons of 511 keV — the rest mass energy of the electron — are released in nearly opposite directions. Detecting both of them in coincidence defines a line of response (LOR) along which the annihilation took place, and the accumulation of millions of LORs allows the three-dimensional distribution of activity to be reconstructed [1]. This detection scheme gives PET a functional sensitivity that few other non-invasive imaging techniques can match [3], at the price of the tracer itself: positron-emitting radionuclides are radioactive by nature, costly to produce, and short-lived, which ties any PET facility to a nearby cyclotron and specific facilities [2].

This functional readout is most valuable where disease alters metabolism before it alters anatomy. In oncology, ^{18}F-FDG PET has become a cornerstone for diagnosis, staging, restaging and the assessment of treatment response [4], [5]. Its role in neurology is arguably even more distinctive, since it reveals pathologies that produce no macroscopic structural abnormality on conventional Magnetic Resonance Imaging (MRI) or Computed Tomography (CT), such as the characteristic hypometabolic and amyloid patterns of Alzheimer's disease [6], among other neurodegenerative and cerebrovascular disorders.

The sensitivity of PET is not matched by its spatial resolution. Clinical scanners typically resolve 4–5 mm mm full width at half maximum (FWHM) [7], and recent advances push the number towards 3 mm [7], [8], higher than the numbers normally achieved with MRI or CT. This limit decomposes into four contributions, two physical and two belonging to the detector, which behave very differently from one another [9]. The physical ones are the finite distance the positron travels before annihilating, set by the emitting isotope [10], and the small angular deviation from antiparallel emission, caused by the residual momentum of the electron–positron pair. Neither can be engineered away, although the second translates into a positional error that scales with the diameter of the detector ring and is therefore smaller in compact systems. The detector terms are more tractable. In pixelated arrays, the size of the individual element fixes the finest scale at which an interaction can be localised; shrinking it improves resolution, but the inter-crystal reflectors then occupy a growing fraction of the detector volume, and the probability that a photon deposits energy across several elements rises. The decoding error — the uncertainty in determining where within the module the interaction occurred — amounts to roughly one third of the detector width [9]. Unlike positron range and non-collinearity, this term is not a physical floor but a consequence of how the light distribution is read and interpreted, and it is therefore the one most open to improvement, whether through detector geometry, readout

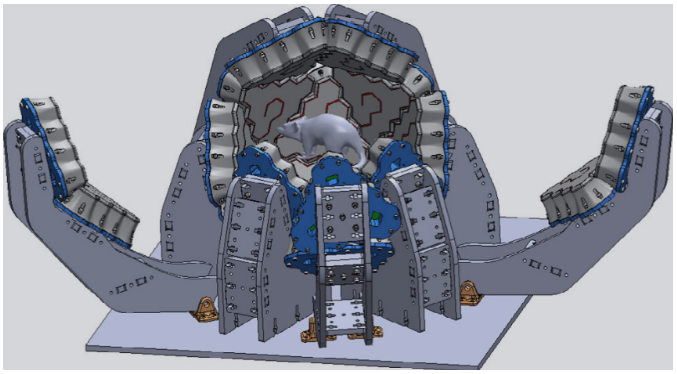

Figure 1: **TEMPORAL**Three-dimensional model of the quasi-spherical, hexagonal-based PET scanner. Reproduced from [12].

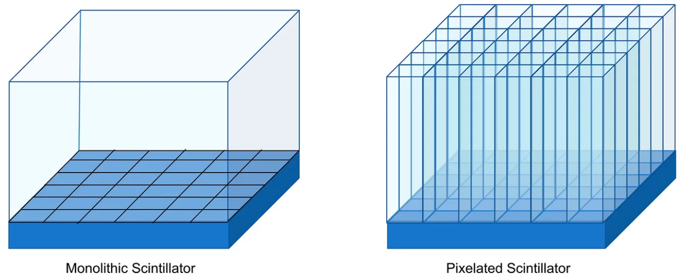

Figure 2: Monolithic (left) versus pixelated (right) scintillator configurations. In the monolithic design a continuous slab is coupled to a segmented photosensor array; in the pixelated design the crystal is cut into elements coupled one-to-one to the photosensors. Reproduced from [17] under CC BY 4.0.

electronics or the algorithms that interpret the signal. This is the ground occupied by organ-dedicated scanners. Adapting the detector geometry to a specific anatomical region increases solid-angle coverage, and with it sensitivity and resolution, at a lower manufacturing cost than a general-purpose whole-body system; designs exist for breast, brain, prostate and cardiac imaging [11]. The trade-off is that irregular geometries complicate calibration and data correction, and that small gantry apertures accentuate parallax errors and degrade the uniformity of spatial resolution across the field of view [11]. Much of the design effort in these systems therefore shifts onto the detector itself.

Our group is currently developing a quasi-spherical brain-dedicated PET scanner along these lines [12]. Rather than the cylindrical ring of conventional systems, its detector modules tile a closed surface around the head (Fig. 1), approaching full 4π steradian coverage and thereby collecting a far larger fraction of the emitted coincidences [12]. The modules are hexagonal, a shape chosen for two independent reasons: hexagons tile a surface without gaps and approximate a sphere more closely than square modules of comparable size, and hexagonal scintillators have also been shown to offer slightly better light yield and energy resolution than square, triangular or cylindrical prisms [13]. Each module is built around a monolithic LYSO crystal [14].

The choice between segmenting the scintillator and leaving it continuous is not incidental, and it conditions how the interaction position can be recovered at all [15]. In pixelated arrays, the crystal is cut into individual elements separated by reflectors, so that the interaction is assigned to whichever element fired. Event positioning is then simple and fast, but the resolution cannot be finer than the element itself [9], and two costs follow from making the elements smaller: the reflectors occupy a growing fraction of the detector volume, reducing the packing fraction, and the number of readout channels rises. Multiplexing schemes, in which several crystals share a readout line and the position is recovered by decoding the light distribution, are commonly used to keep this cost manageable [11]. Pixelated arrays also provide no intrinsic depth-of-interaction (DOI) information, which has motivated layered designs such as phoswich detectors, where stacked scintillators with different decay times allow a coarse depth assignment at the price of a degraded timing resolution [11].

Monolithic scintillators take the opposite approach. A single continuous block is coupled to an array of photodetectors, and the light from each interaction spreads over many channels at once [15]. Positioning is no longer a matter of identifying an element but of estimating a coordinate from the recorded light distribution, which shifts the resolution limit from the geometry of the crystal to the quality of the positioning algorithm, and makes the DOI recoverable as a continuous variable [16]. The absence of internal reflectors also restores the packing fraction lost in segmented designs [11]. This is the configuration adopted in the scanner described above, and it is the light sharing across the array that the present work exploits.

This light distribution is read out by an array of silicon photomultipliers (SiPMs) optically coupled to the rear face of the crystal [15]. Each interaction produces signal in a subset of channels, and a readout application-specific integrated circuit (ASIC) amplifies, thresholds and digitises each of them, so that an event is fully described by the vector of charges recorded across the array. Because a channel only produces an output once it exceeds the threshold set in the electronics, most entries of that vector are zero for any given event and only a few channels contribute; the sparsity of the signal is therefore a property of the acquisition system rather than a defect of the data [18].

Estimating the interaction position from that vector admits solutions of widely varying complexity. The most common is the centre of gravity (CoG), which places the event at the average of the photodetector coordinates, each weighted by the charge that channel recorded [19]; in practice the square of the charge is normally used as the weight, which reinforces the contribution of the channels closest to the interaction point and improves resolution [20]. Its appeal is that it is deterministic, has no parameters to tune and requires no prior calibration. In exchange, it exploits only part of the information contained in the light distribution: recovering the DOI, or correcting the

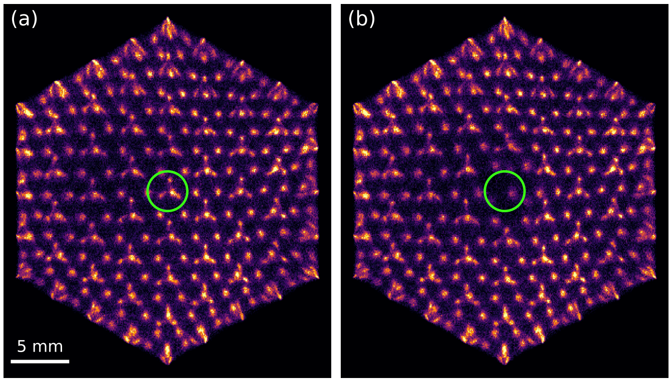

Figure 3: Flood map of the same detector module reconstructed by centre of gravity, using the same events in both panels: (a) with all 61 channels operational and (b) with one channel disabled, marked in green. The structures engraved in the crystal appear as discrete spots; the inset magnifies the neighbourhood of the disabled channel.

bias the method itself introduces near the crystal edges, calls for substantially more elaborate algorithms [21].

The two-dimensional histogram of the positions computed in this way over a large number of events forms the flood map (Fig. 3), in which the structures engraved in the crystal appear resolved as discrete spots. It is the image conventionally used to characterise a detector of this kind, and its sharpness depends on all 61 channels being operational: the CoG averages over the whole array, and a single missing term is enough to displace the centroid. The effect is confined to the neighbourhood of the failing channel — in the module of Fig. 3 the reconstructed position shifts by a median of 0.41 mm for events within 4 mm of it, and by less than 0.01 mm elsewhere — but within that neighbourhood the engraved structures are no longer resolved.

Silicon photomultipliers degrade and fail for a variety of reasons: thermal drift, manufacturing defects, deterioration of the optical coupling, or faults in the readout channel. The consequence for the image is well recognised in industrial practice: a dead pixel introduces reconstruction artefacts and incorrect activity values, and commercial systems include quality-control procedures to identify it [22]. The adopted general solution, however, is to discard the affected information and compensate for it downstream of positioning, either by duplicating events from neighbouring pixels in list mode or by interpolating in sinogram space [22]. The channel is not recovered: it is masked out.

This class of problem has motivated the application of deep learning to PET detector readout [23]. The most developed line is the estimation of the interaction position in monolithic crystals, where networks trained on calibration data consistently outperform analytical algorithms and recover the DOI as a continuous variable [24], [25]. Two previous studies address precisely this task on the present detector [18], [26]. All of them share the same framing: the light distribution recorded by the array is taken as input and the network returns a coordinate.

Closer to what is proposed here, deep learning has also been applied to recovering channel signals rather than positions. Networks have been trained to invert multiplexed readout schemes, reconstructing the individual channel values from a reduced set of measurements and validating the result on the recovered flood map [27], [28], and the approach has recently been extended to semi-monolithic detectors, where the full 64-channel response is reconstructed from 16 row-and-column sums by exploiting the light sharing between slabs [29]. In all these cases, however, the information carried by each channel is still present in the acquired data: multiplexing encodes it in a known linear combination, and the network learns to invert a mapping fixed by design. A failed channel is a different situation, since nothing of its signal reaches the acquisition and the only remaining trace of it is the statistical correlation that light sharing induces among its neighbours.

That situation is not unique to PET. Recovering the response of defective elements in a sensor array is an established problem in imaging, addressed either by interpolating from neighbouring pixels or, more recently, by training autoencoders to reconstruct corrupted regions from their surroundings [30]. The formulation adopted here belongs to that family: it is the masked autoencoder setting [31], in which part of the input is deliberately hidden and the model is trained to reconstruct it, which allows supervision to be generated from healthy modules without requiring labelled faulty detectors. What distinguishes the present case is that the redundancy being exploited is physical rather than statistical: the light emitted in a single interaction is measured simultaneously by several photodetectors, so the value of a channel is constrained by the response of those around it.

This work proposes to recover that channel rather than discard it. If the light from each interaction spreads over several photodetectors, the charge that the failed sensor would have recorded is not entirely lost: it remains implicit in what its neighbours measured, and the problem is to estimate it. The difficulty lies less in the model than in the supervision. In a faulty module there is no way of checking whether an estimate is correct, since the true charge of the dead channel is unobservable by definition: if it could be measured, the channel would not be faulty. What cannot be observed, however, can be simulated. It suffices to take a healthy module, disable one of its channels and keep the original value as a reference; the fault is fabricated, and with it the label. The whole approach rests on that idea, and it is applied to experimental data acquired with the detector described above rather than to simulations, so that the model learns the behaviour of the system as it actually is, with its noise, its threshold and its variation between modules.

The choice of model follows from the geometry of the array. The immediate option would be to treat the vector of sixty-one charges as a list of numbers and let a dense network discover on its own which channels are related to which, but that means learning from scratch something that is already known: which sensor is adjacent to which. The photodetectors form a hexagonal tiling in which every interior sensor has six

Table I: Resumen de hiperparametros.

Property	**Model A**	**Model B**
Epochs	XX	XX
Learning rate	XX	XX
Batch size	XX	XX

immediate neighbours, and that neighbourhood is precisely the path along which the light is shared. A structure of this kind is naturally described as a graph, with one node per sensor and an edge between each pair of adjacent sensors, and it therefore admits a neural network that operates on it by propagating information between neighbours [32]. The model proposed here follows that approach, incorporating the geometry of the detector as prior knowledge instead of leaving it implicit in the data.

The evaluation is framed along the same lines. The error in the estimated charge is the quantity the model minimises, but on its own it says little about the usefulness of the method: what matters is how much is recovered of what the fault degrades. Two physical effects are therefore measured: the displacement of the reconstructed position, which is the direct consequence on the flood map, and the conservation of the energy spectrum, which determines whether the restored signal is physically admissible. All of this on modules the network has not seen during training, and compared against classical interpolation methods in order to establish that the complexity of the model is warranted. The behaviour of the method when more than one channel fails is also examined, with particular attention to contiguous groups, which are what is observed in real faulty detectors and the case that displaces the reconstructed position the most, since when an entire region disappears the lost sensors are left without healthy immediate neighbours to draw on.

II. Materials and Methods

A. Dataset and Experimental Setup

Origen de los datos, tamaño de muestra, particiones train/val/test.

B. Preprocessing

$$x_{i,j} = \frac{x_{i,j} - \min(\mathbf{x}_i)}{\max(\mathbf{x}_i) - \min(\mathbf{x}_i) + \varepsilon} \tag{1}$$

C. Proposed Model

Arquitectura, hiperparametros y entrenamiento.

D. Performance Evaluation

Metricas empleadas.

III. Results

Comenta la Fig. 4 y la Tabla I.

IV. Discussion

Interpretacion, comparacion con el estado del arte y limitaciones.

Figure 4: Figura de una columna.

V. Conclusions

Conclusiones y lineas futuras.

References

[1] S. R. Cherry, J. A. Sorenson, and M. E. Phelps, *Physics in Nuclear Medicine*, 4th ed. Philadelphia, PA: Elsevier Saunders, 2012.

[2] J. Rong, A. Haider, T. E. Jeppesen, L. Josephson, and S. H. Liang, "Radiochemistry for positron emission tomography," *Nature Communications*, vol. 14, 2023.

[3] J. J. Vaquero and P. Kinahan, "Positron emission tomography: Current challenges and opportunities for technological advances in clinical and preclinical imaging systems," *Annual Review of Biomedical Engineering*, vol. 17, pp. 385–414, 2015.

[4] J. W. Fletcher, B. Djulbegovic, H. P. Soares *et al.*, "Recommendations on the use of ^{18}f-fdg PET in oncology," *Journal of Nuclear Medicine*, vol. 49, no. 3, pp. 480–508, 2008.

[5] R. Boellaard, R. Delgado-Bolton, W. J. G. Oyen *et al.*, "FDG PET/CT: EANM procedure guidelines for tumour imaging: Version 2.0," *European Journal of Nuclear Medicine and Molecular Imaging*, vol. 42, no. 2, pp. 328–354, 2015.

[6] K. A. Johnson, S. Minoshima, N. I. Bohnen *et al.*, "Appropriate use criteria for amyloid PET," *Journal of Nuclear Medicine*, vol. 54, no. 3, pp. 476–490, 2013.

[7] B. A. Spencer, E. Berg, J. P. Schmall *et al.*, "Performance evaluation of the uEXPLORER total-body PET/CT scanner based on NEMA NU 2-2018," *Journal of Nuclear Medicine*, vol. 62, no. 6, pp. 861–870, 2021.

[8] A. González-Montoro, J. Barberá, D. Sánchez, Á. Mondéjar, M. Freire, K. Díaz, A. Lucero, S. Jiménez-Serrano, J. Alamo, C. Morera-Ballester, J. Barrio, N. Cucarella, V. Ilisie, L. Moliner, C. Valladares, A. J. González, J. Prior, and J. M. Benlloch, "A new brain dedicated PET scanner with 4D detector information," *Bio-Algorithms and Med-Systems*, vol. 18, no. 1, pp. 107–119, 2022.

[9] W. W. Moses, "Fundamental limits of spatial resolution in PET," *Nuclear Instruments and Methods in Physics Research A*, vol. 648, no. Supplement 1, pp. S236–S240, 2011.

[10] C. S. Levin and E. J. Hoffman, "Calculation of positron range and its effect on the fundamental limit of PET system spatial resolution," *Physics in Medicine and Biology*, vol. 44, no. 3, pp. 781–799, 1999.

[11] A. Sanaat, M. Amini, H. Arabi, and H. Zaidi, "The quest for multifunctional and dedicated PET instrumentation with irregular geometries," *Annals of Nuclear Medicine*, vol. 38, no. 1, pp. 31–70, 2024.

[12] D. Pérez-Benito, R. Chil, J. M. Udías, M. Desco, and J. J. Vaquero, "SiPM-based PET detector module for a 4π span scanner," vol. 936, pp. 18–21.

[13] D. Perez-Benito, R. Chil, L. A. Hidalgo-Torres, and J. J. Vaquero, "Scintillator geometrical considerations for detectors based on hexagonal SiPMs," vol. 7, no. 7, pp. 684–691. [Online]. Available: https://ieeexplore.ieee.org/document/10133867/

[14] D. Perez-Benito, L. Batres, M. Vidal, and J. J. Vaquero, "Laser-microstructured hexagonal detectors for high-sensitivity quasi-spherical brain PET," in *2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD)*, pp. 1–1. [Online]. Available: https://ieeexplore.ieee.org/document/11287358/

[15] A. González-Montoro, M. N. Ullah, and C. S. Levin, "Advances in detector instrumentation for PET," *Journal of Nuclear Medicine*, vol. 63, no. 8, pp. 1138–1144, 2022.

[16] S. España, R. Marcinkowski, V. Keereman, S. Vandenberghe, and R. Van Holen, "DigiPET: sub-millimeter spatial resolution small-animal PET imaging using thin monolithic scintillators," vol. 59, pp. 3405–3420.

[17] E. Enlow and S. Abbaszadeh, "State-of-the-art challenges and emerging technologies in radiation detection for nuclear medicine imaging: A review," *Frontiers in Physics*, vol. 11, p. 1106546, 2023.

[18] L. Jiménez Algarra, "Deep learning for precise gamma-ray interaction in a hexagonal monolithic PET detector," Master's thesis, Universidad Carlos III de Madrid, Leganés, Spain, 2025.

[19] H. O. Anger, "Scintillation camera," vol. 29, no. 1, pp. 27–33.

[20] H. Poladyan, O. Bubon, A. Teymurazyan, S. Senchurov, and A. Reznik, "Gaussian position-weighted center of gravity algorithm for multiplexed readout," vol. 65, no. 16, p. 165003.

[21] P. Conde, A. Iborra, A. J. González, L. Hernández, P. Bellido, E. Crespo, L. Moliner, J. P. Rigla, M. J. Rodríguez-Álvarez, F. Sánchez, M. Seimetz, A. Soriano, L. F. Vidal, and J. M. Benlloch, "Minimization of border effects in monolithic scintillators using neural networks, based on MR-compatible SiPM arrays," vol. 1, no. 1, p. A19. [Online]. Available: https://doi.org/10.1186/2197-7364-1-S1-A19

[22] T. L. Laurence and S. X. Wang, "Dead pixel identification in positron emission tomography (PET)," U.S. Patent 9,841,515 B2 (WO 2015/145310 A1), Koninklijke Philips N.V., 2017, prioridad 28 mar. 2014.

[23] K. Gong, E. Berg, S. R. Cherry, and J. Qi, "Machine learning in PET: From photon detection to quantitative image reconstruction," *Proceedings of the IEEE*, vol. 108, no. 1, pp. 51–68, 2020.

[24] M. Decuyper, M. Stockhoff, S. Vandenberghe, and R. Van Holen, "Artificial neural networks for positioning of gamma interactions in monolithic PET detectors," *Physics in Medicine and Biology*, vol. 66, no. 7, p. 075001, 2021.

[25] G. Jaliparthi, P. F. Martone, A. V. Stolin, and R. R. Raylman, "Deep residual-convolutional neural networks for event positioning in a monolithic annular PET scanner," vol. 66, no. 14, p. 145008.

[26] L. A. Hidalgo-Torres, D. Pérez-Benito, R. Chil, P. Macias, and J. J. Vaquero, "Predicting 3D photon interaction in a hexagonal positron emission tomography detector: A deep learning approach," in *CASEIB 2020, XXXVIII Congreso Anual de la Sociedad Española de Ingeniería Biomédica: libro de actas*, R. Hornero Sánchez and Sociedad Española de Ingeniería Biomédica, Eds. Universidad de Valladolid, faltan las paginas: sacalas del PDF del libro de actas.

[27] Y. Li, X. Cheng, A. LaBella, W. Zhao, and A. H. Goldan, "Interleaved signal multiplexing readout in depth-encoding Prism-PET detectors," *Medical Physics*, vol. 50, no. 9, pp. 5327–5338, 2023, demultiplexado con DL evaluado por recuperacion del flood map.

[28] M. Niu, X. Jiang, S. Huang, Z. Chen, Y. Zhao, Y. Yang, W. Huang, H. Yang, X. Zhang, Z. Deng, and Z. Gu, *A CNN-based demultiplexing method for an interleaved signal multiplexing readout with a TOF-DOI brain PET detector*, pages: 1.

[29] F. Loignon-Houle, F. Pagano, and A. J. Gonzalez, "Multi-input interaction positioning and signal demultiplexing with deep learning in semi-monolithic detectors," vol. 10, no. 4, pp. 547–557. [Online]. Available: https://ieeexplore.ieee.org/document/11177553/

[30] S. Sarkar, X. Ye, G. Datta, and P. A. Beerel, "Fixpix: Fixing bad pixels using deep learning," 2024. [Online]. Available: https://arxiv.org/abs/2310.11637

[31] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 16 000–16 009.

[32] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, pp. 1024–1034.